

Exploration of α -conjugate Decays Pathways

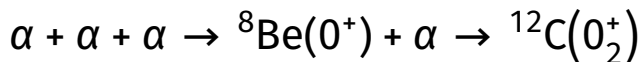
Bryan M Harvey, December 3, 2025

Interplay of Structure and Reaction Rates

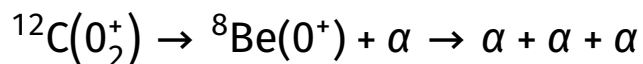
Structurally:

- ${}^8\text{Be}(0^+)$ is very similar to two connected α s
- ${}^{12}\text{C}(0_2^+)$ is very similar to an α connected to ${}^8\text{Be}(0^+)$

Triple- α Process

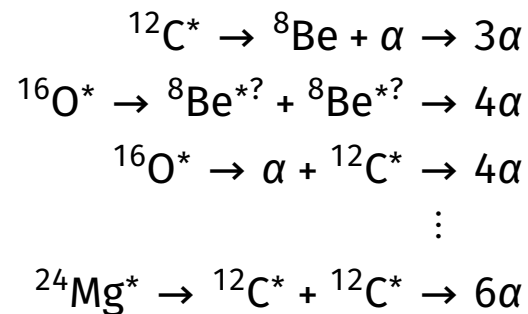


Studied via



Models now predict α -clusterization in nuclei as large as calcium

- Can we see evidence of previously unexplored α -conjugate decay pathways in excited light-medium sized nuclei?



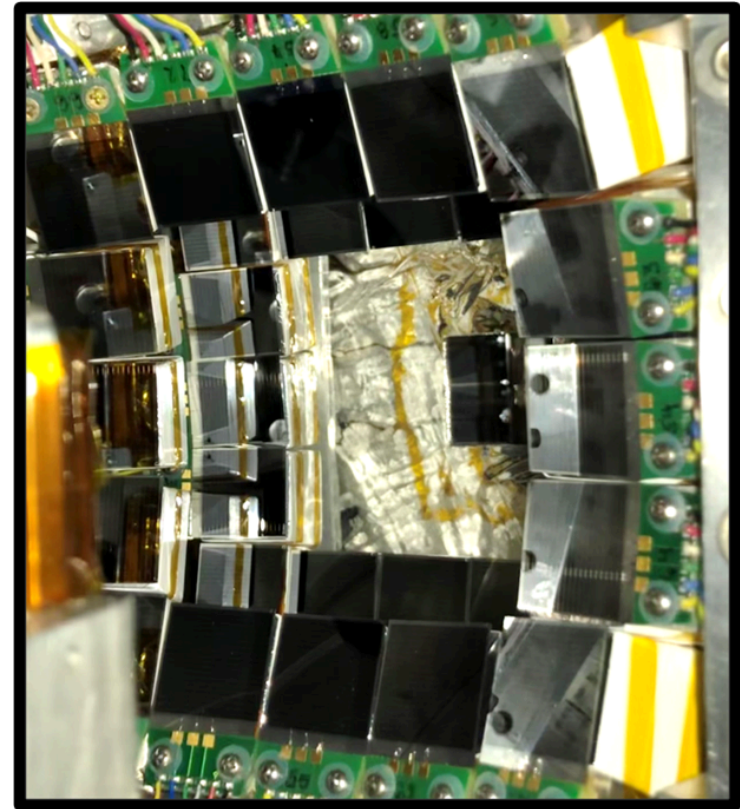
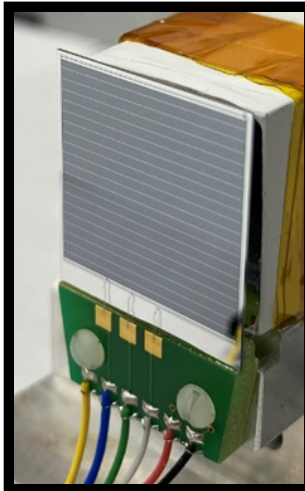
Experimental Design

System

- $^{16}\text{O} + \text{natC}$ @ 35 MeV/u
 - Also have ^{20}Ne , ^{24}Mg , ^{28}Si , ^{32}S , ^{36}Ar projectiles

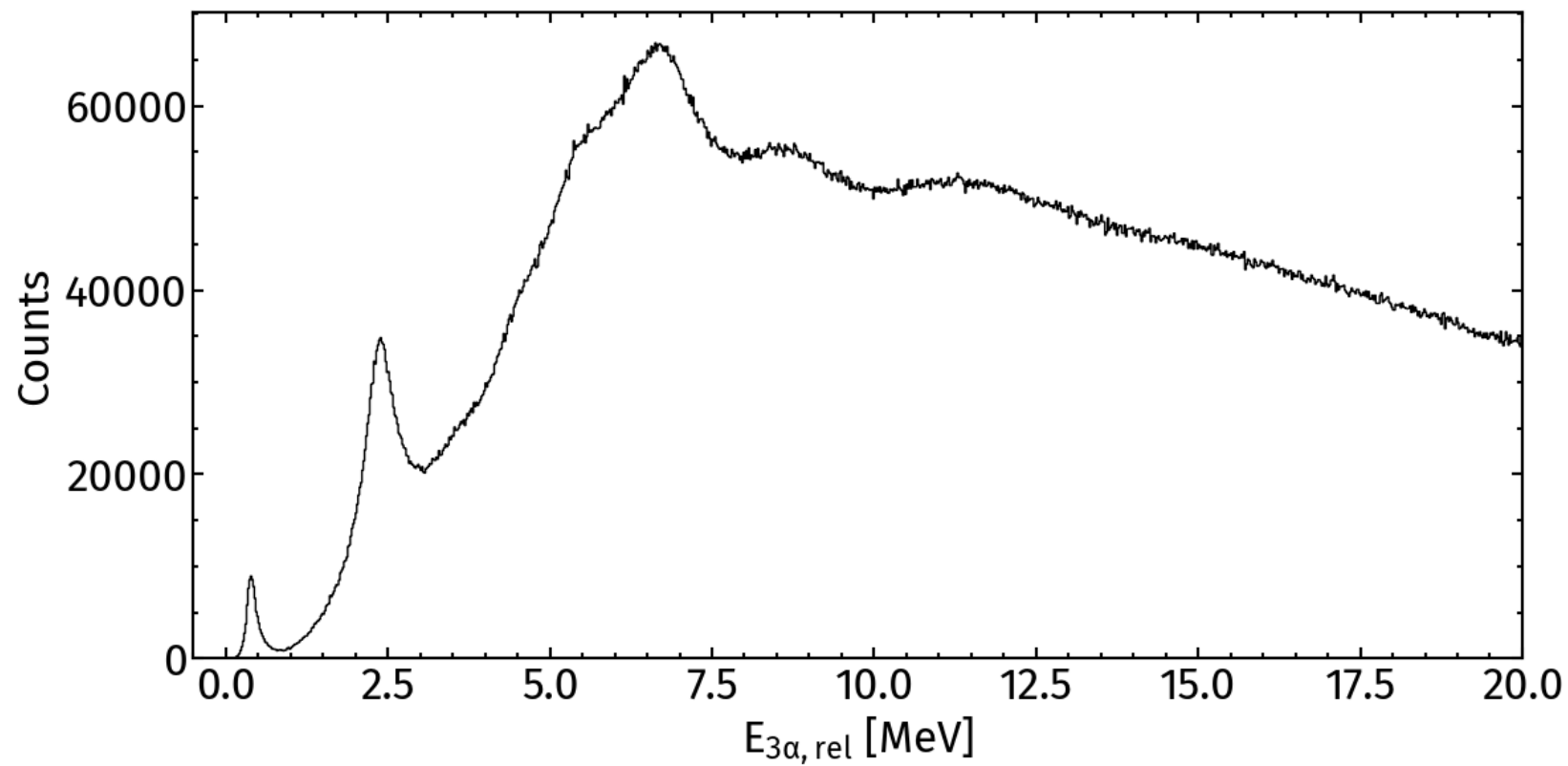
Forward Array Using Silicon Technology (FAUST)

- Charged Particle Identification
 - $(\Delta E, E) \rightarrow (Z, A)$
- Total Kinetic Energy
 - Particles stop in the thick CsI(Tl)
- Position Sensitivity
 - Achieve sub-mm position uncertainty

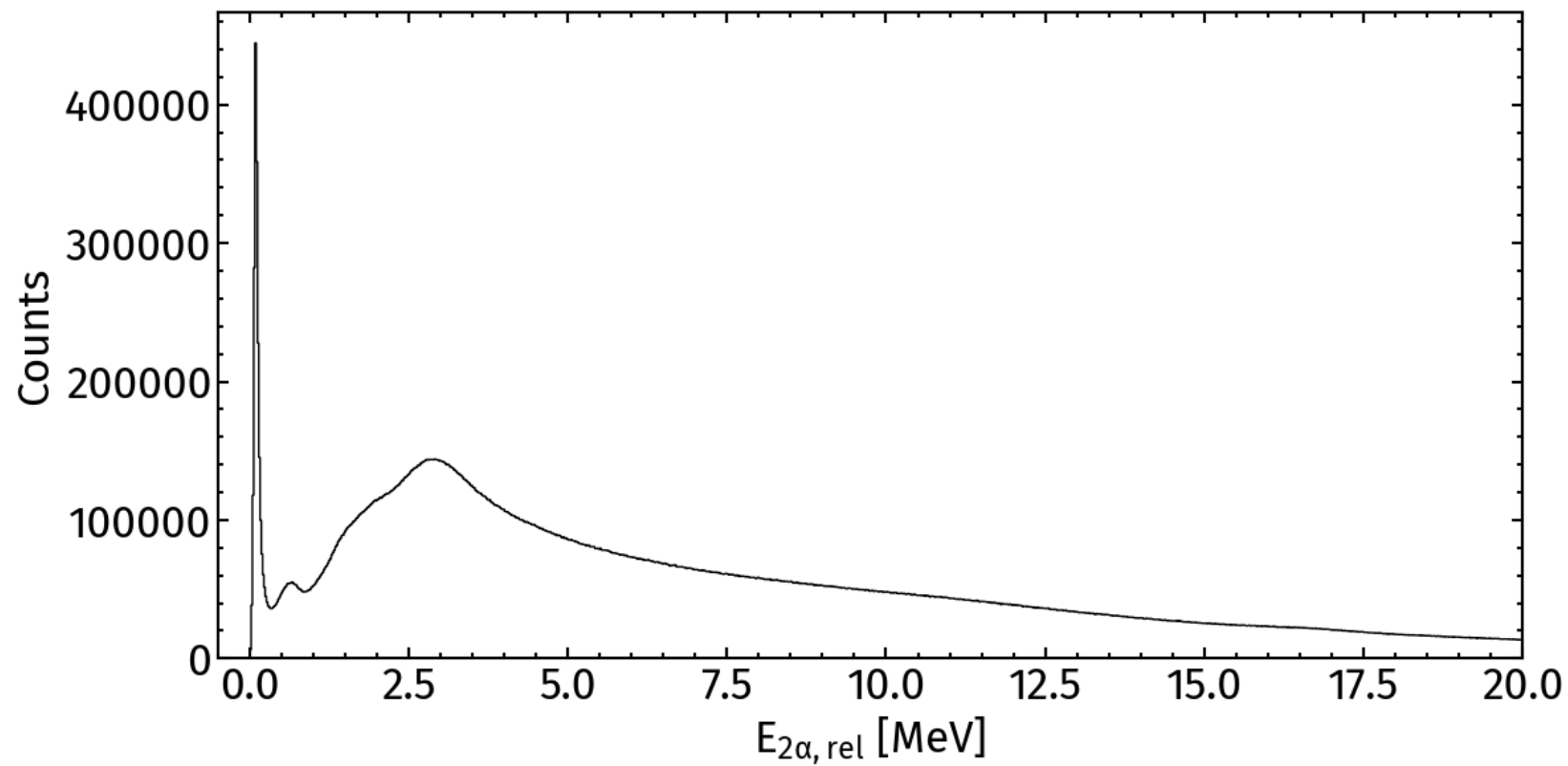


$$E_{\text{rel}} = \sum_i T_{i,\text{c.m.}} = E^* + Q_{\text{rxn}}$$

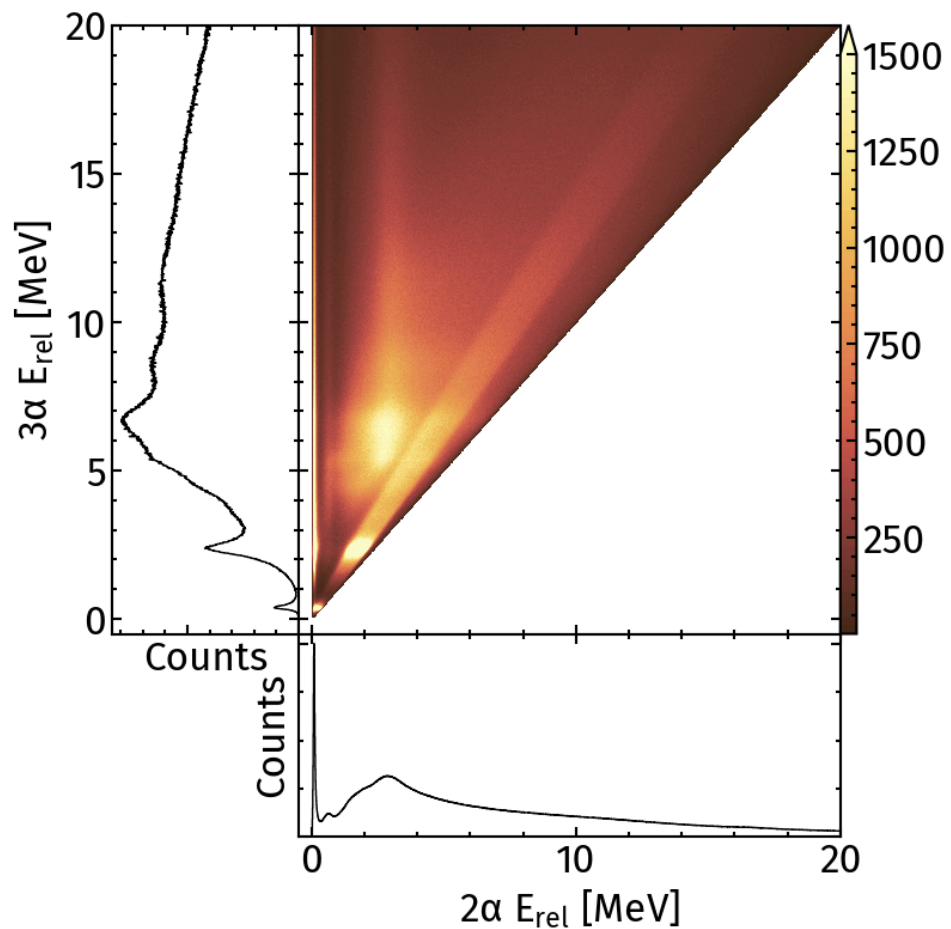
$E_{3\alpha, \text{rel}}$ (FIXME: ADD LABEL ASSETS FROM WPCF)



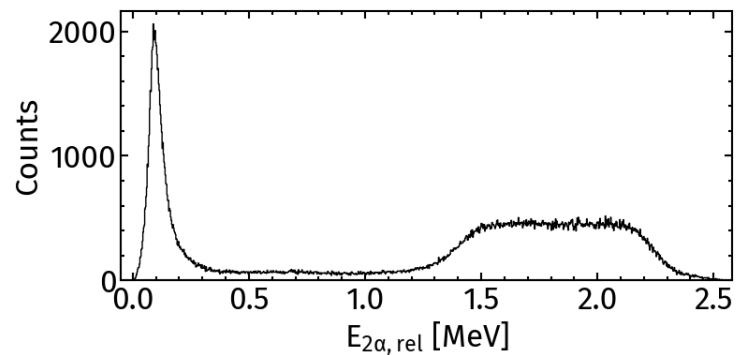
$E_{2\alpha, \text{rel}}$ (FIXME: ADD LABEL ASSETS FROM WPCF)



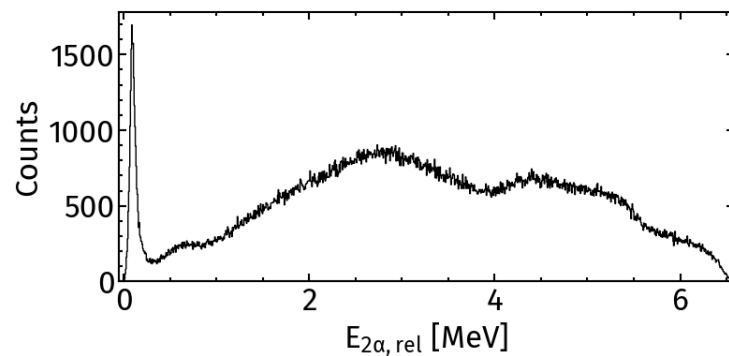
$E_{3\alpha, \text{rel}}$ versus $E_{2\alpha, \text{rel}}$



$$E_{3\alpha, \text{rel}} = 2.5 \text{ MeV} \pm 80 \text{ keV}$$

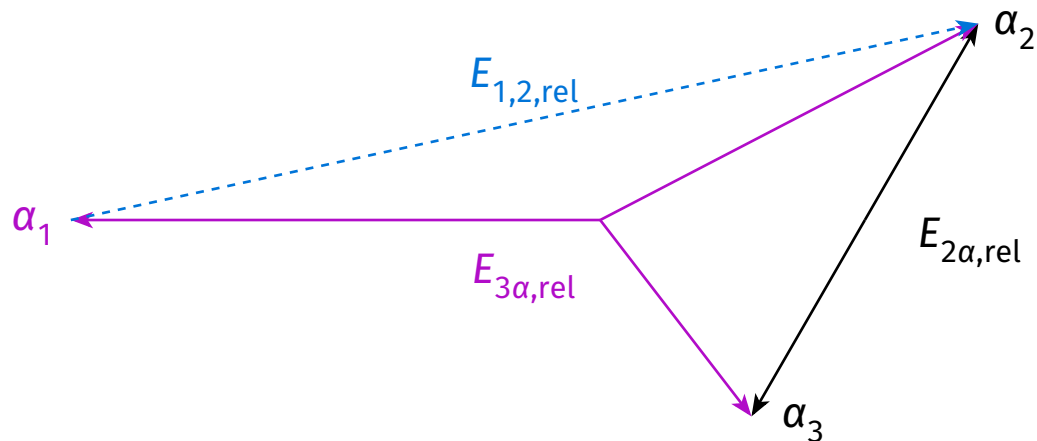
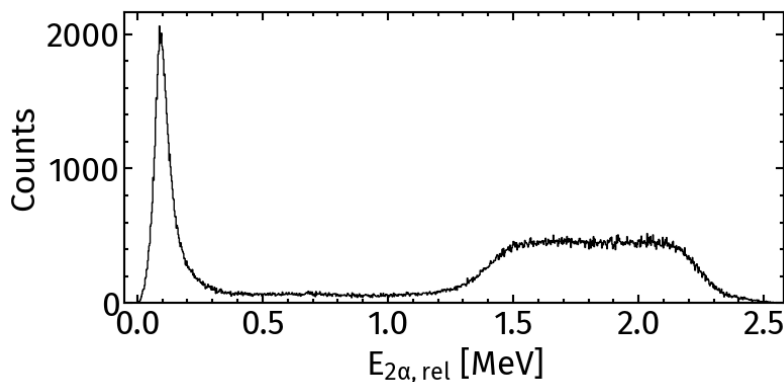


$$E_{3\alpha, \text{rel}} = 6.5 \text{ MeV} \pm 80 \text{ keV}$$



Kinematics (AUDIENCE, HELP WITH FLOW PLEASE)

$$E_{3\alpha, \text{rel}} = 2.5 \text{ MeV} \pm 80 \text{ keV}$$



- Total Spectrum (Measured):

$$T(E_{2\alpha, \text{rel}}) = U(E_{2\alpha, \text{rel}}) + S(E_{2\alpha, \text{rel}}) + \hat{K}[S(E_{2\alpha, \text{rel}})]$$

- U – Uncorrelated (no ^8Be)
- S – Signal (^8Be)

- $\hat{K}[S]$ – Kinematically constrained contribution due to S .
 - Given a 3α E_{rel} and a real ^8Be 2α E_{rel} , what is the distribution of $\text{non-}^8\text{Be}$ 2α E_{rel} ?

Solution without proof

$$\hat{K}[S(E_{2\alpha, \text{rel}})] = 2 \int_{E_-}^{E_+} \frac{S(x)dx}{E_+ - E_-}$$

where

$$E_{\pm}(E_{2\alpha, \text{rel}}; E_{3\alpha, \text{rel}}) = \frac{3}{4}E_{3\alpha, \text{rel}} - \frac{1}{2}E_{2\alpha, \text{rel}} \pm \frac{1}{2}\sqrt{3E_{2\alpha, \text{rel}}(E_{3\alpha, \text{rel}} - E_{2\alpha, \text{rel}})}$$

Integral Equation to solve for S :

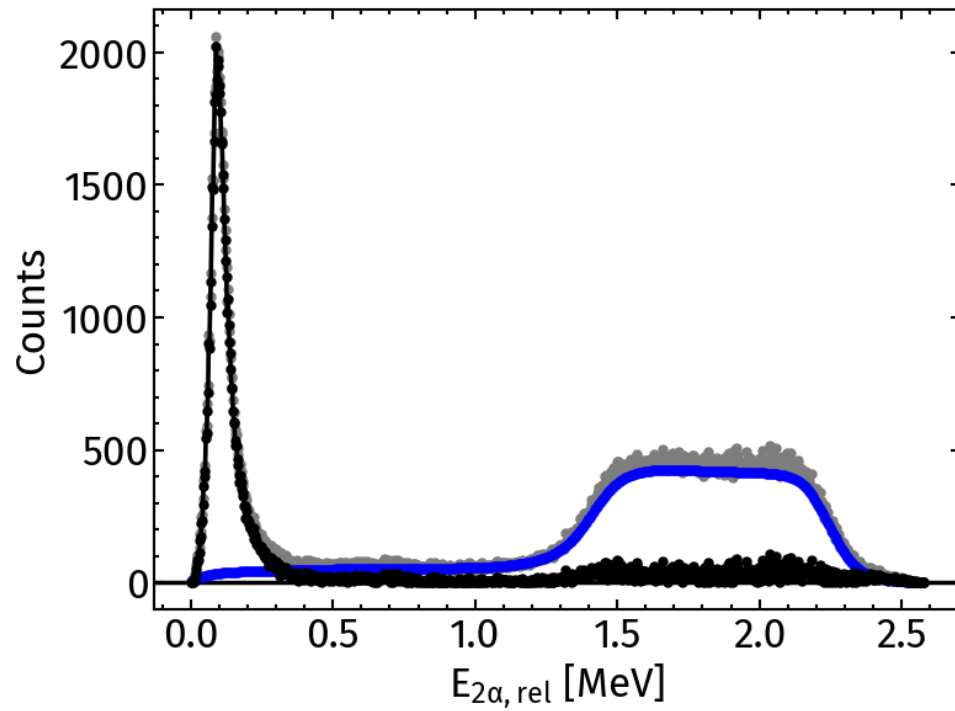
$$T(E_{2\alpha, \text{rel}}) - U(E_{2\alpha, \text{rel}}) = (1 + \hat{K})S(E_{2\alpha, \text{rel}})$$

This is numerically solvable if U is known.

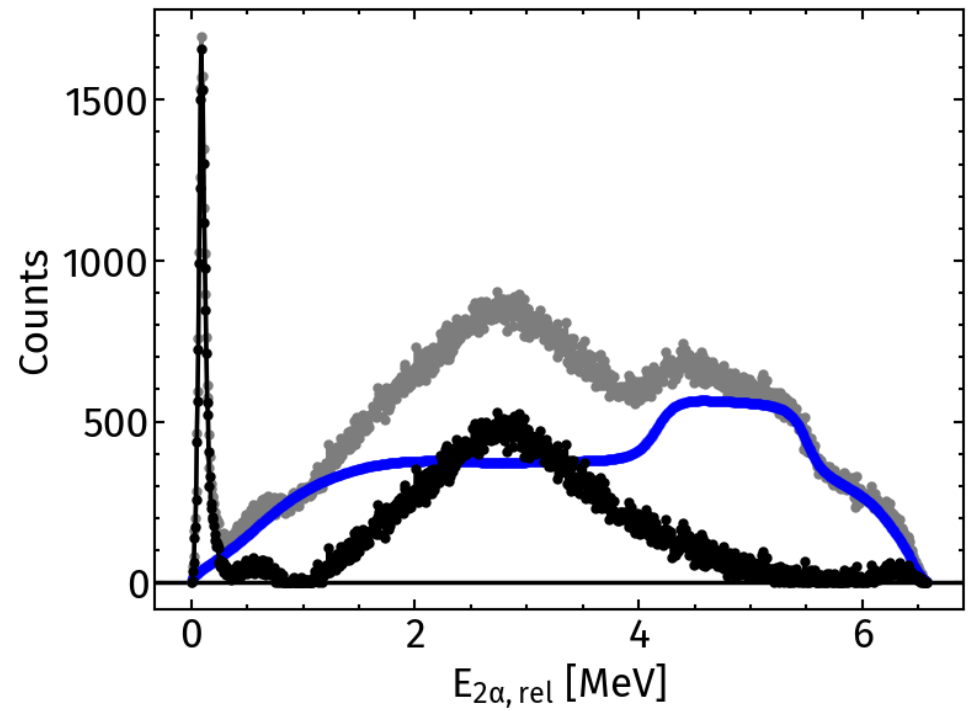
Make the (bad) assumption that $U = 0$

$E_{2\alpha, \text{rel}}$ Decomposition

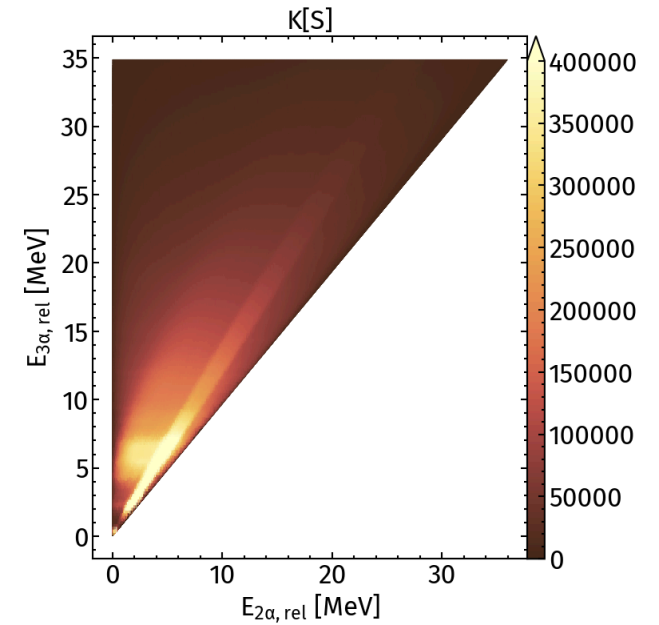
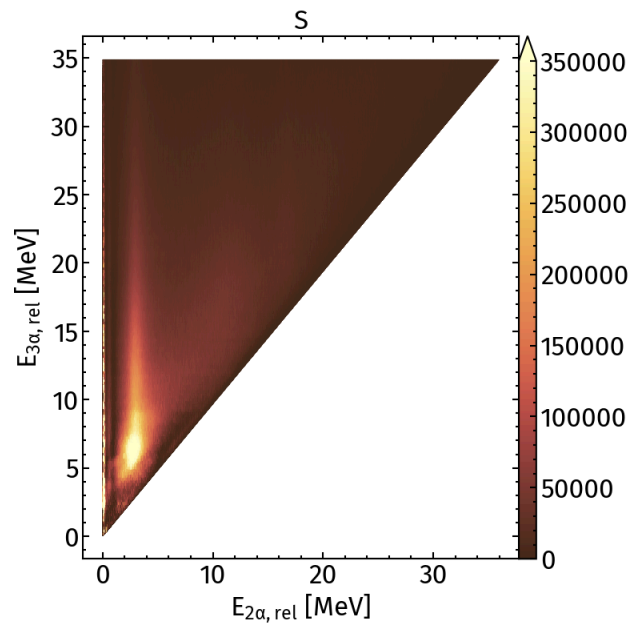
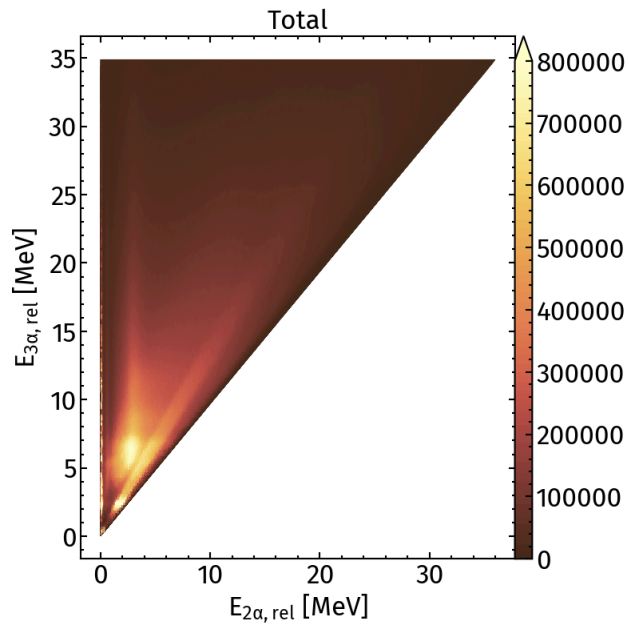
$$E_{3\alpha, \text{rel}} = 2.5 \text{ MeV} \pm 80 \text{ keV}$$



$$E_{3\alpha, \text{rel}} = 6.5 \text{ MeV} \pm 80 \text{ keV}$$



$E_{3\alpha, \text{rel}}$ versus $E_{2\alpha, \text{rel}}$ Decomposition



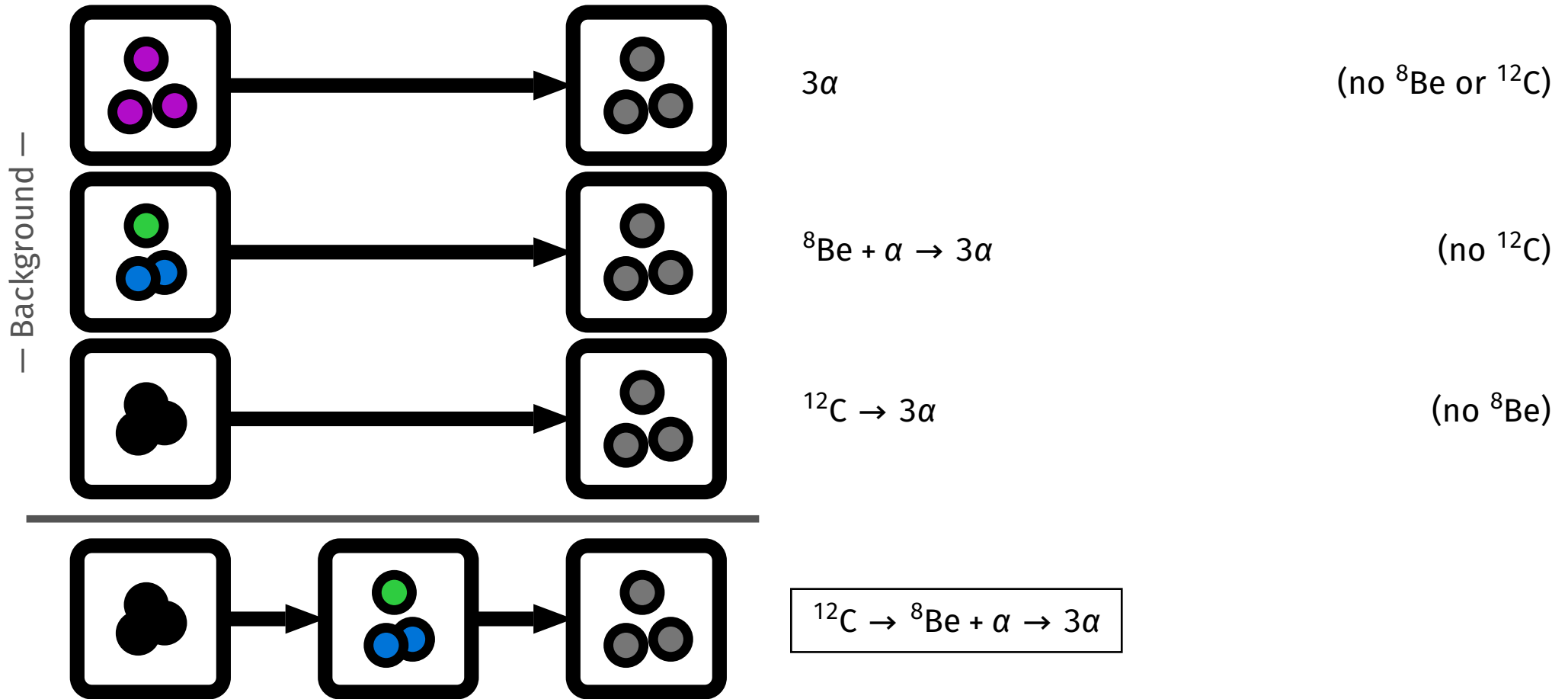
TAPIR (??)

Timeline

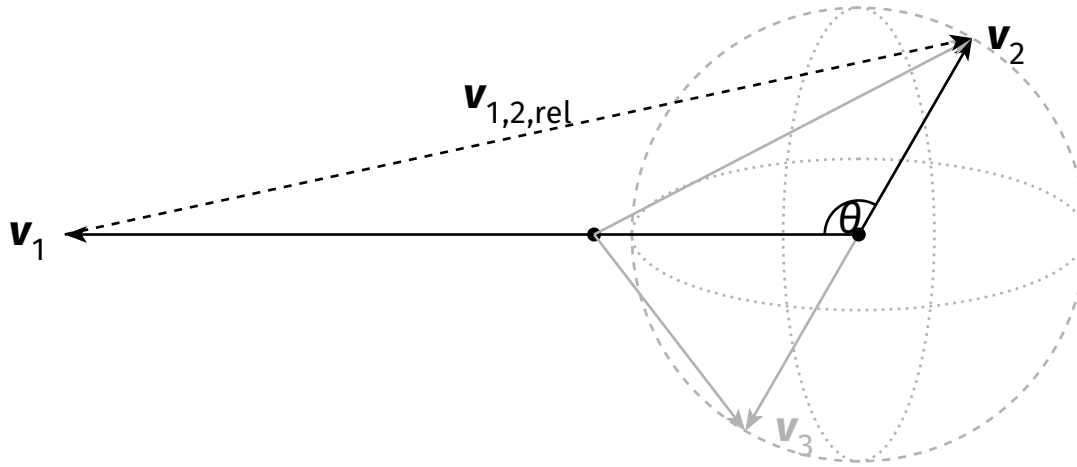
Acknowledgements

Extras

Sources of Background



Solve through velocities



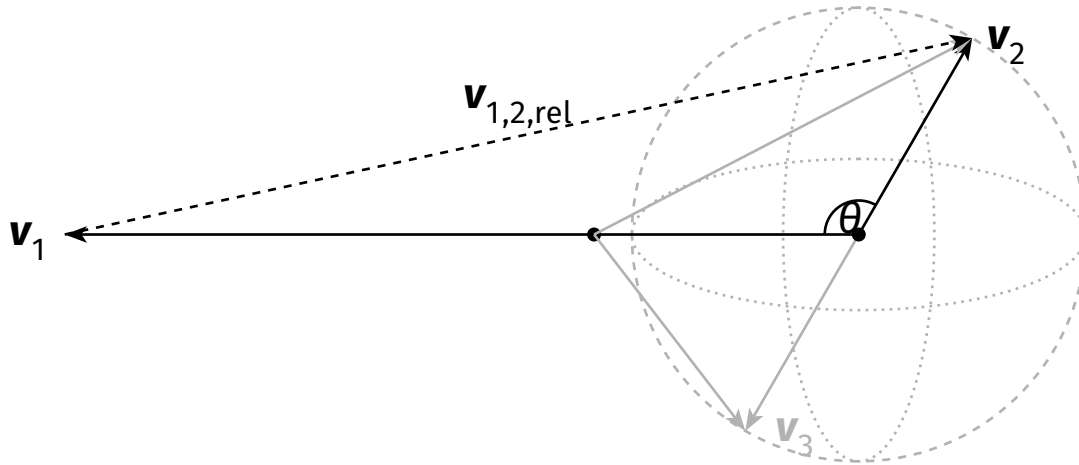
$$v_{1,2,\text{rel}}^2 = v_1^2 + v_2^2 - 2v_1v_2 \cos \theta$$

For isotropic sampling, $\cos \theta$ is sampled uniformly.

$$E_{1,2,\text{rel}} = 2 \left(\frac{1}{2} m_\alpha \left(\frac{v_{1,2,\text{rel}}}{2} \right)^2 \right)$$

$$v_{1,2,\text{rel}}^2 = 4E_{1,2,\text{rel}}/m_\alpha$$

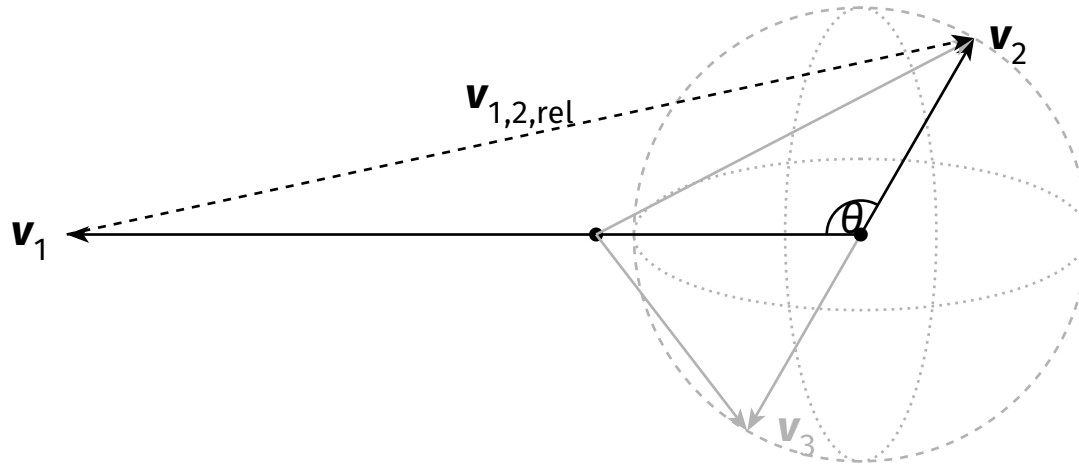
Solve through velocities



$$E_{2\alpha,rel} = 2\left(\frac{1}{2}m_{\alpha}v_{1,2,rel}^2\right)$$

$$v_2 = \sqrt{E_{1,2,rel}/m_{\alpha}}$$

Solve through velocities



In ^{12}C -frame:

$$\begin{aligned} E_{3\alpha,\text{rel}} - E_{2\alpha,\text{rel}} &= \frac{p_{\alpha}^2}{2m_{\alpha}} + \frac{p_{8\text{Be}}^2}{2m_{8\text{Be}}} \\ &\approx \frac{p_{\alpha}^2}{2m_{\alpha}} + \frac{p_{\alpha}^2}{4m_{\alpha}} \\ &= \frac{3}{4}m_{\alpha}v_1^2 \end{aligned}$$

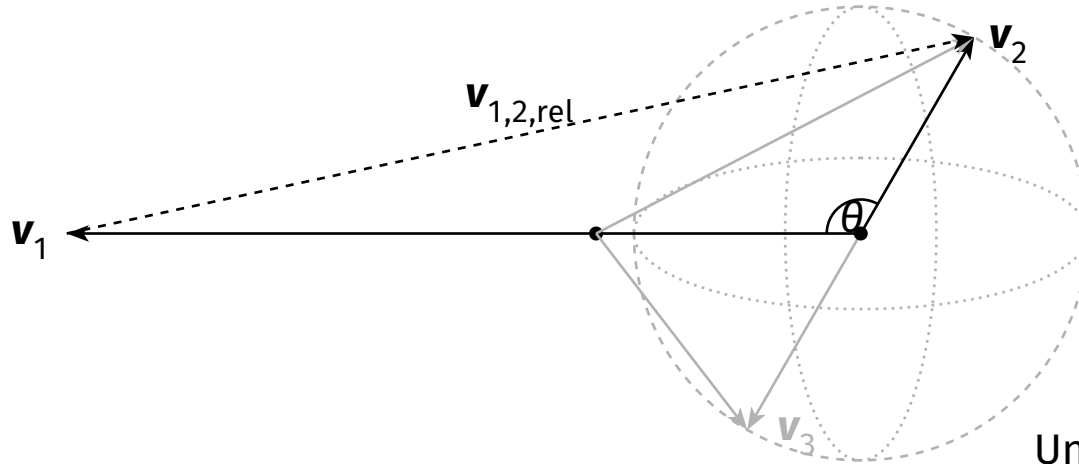
$$v_1 = \sqrt{\frac{4}{3m_{\alpha}}(E_{3\alpha,\text{rel}} - E_{2\alpha,\text{rel}})}$$

Scale by $\frac{3}{2}$ to get to ^8Be -frame

$$v_1 = \sqrt{\frac{3}{m_{\alpha}}(E_{3\alpha,\text{rel}} - E_{2\alpha,\text{rel}})}$$

Solve through velocities

$$v_{1,2,\text{rel}}^2 = v_1^2 + v_2^2 - 2v_1 v_2 \cos \theta$$



$$E_{1,2,\text{rel}} = \frac{3}{4}E_{3\alpha,\text{rel}} - \frac{1}{2}E_{2\alpha,\text{rel}} - \frac{1}{2}\sqrt{3E_{2\alpha,\text{rel}}(E_{3\alpha,\text{rel}} - E_{2\alpha,\text{rel}})}\cos \theta$$

Uniform Distribution with bounds $E_{\pm}$:

$$E_{\pm} = \frac{3}{4}E_{3\alpha,\text{rel}} - \frac{1}{2}E_{2\alpha,\text{rel}} \pm \frac{1}{2}\sqrt{3E_{2\alpha,\text{rel}}(E_{3\alpha,\text{rel}} - E_{2\alpha,\text{rel}})}$$

Solve through velocities

For a given signal $S(E_{2\alpha, \text{rel}})$, there is also a contribution of

$$\hat{K}[S(E_{2\alpha, \text{rel}})] = 2 \int_{E_-}^{E_+} \frac{S(x)dx}{E_+ - E_-}$$

If $T - U$ is known, then:

$$T(E_{2\alpha, \text{rel}}) = U(E_{2\alpha, \text{rel}}) + S(E_{2\alpha, \text{rel}}) + \hat{K}[S(E_{2\alpha, \text{rel}})]$$

$$T(E_{2\alpha, \text{rel}}) - U(E_{2\alpha, \text{rel}}) = (1 + \hat{K})S(E_{2\alpha, \text{rel}})$$

is numerically solvable